Can We Know Snow? Predicting Snowfall in Western Montana

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Date: December 18, 2024

Abstract

This project aims to investigate methods to model snowfall data. Being able to model this data is crucial to understanding the effects of climate change. Our analysis focuses on Western Montana, which has a varied topography. The elevation changes mean that there is significant variation in the snow pack. We utilize a Montana county shapefile, a dataset containing data on more than 150 weather stations in Montana, and a Montana digital elevation model (DEM). Given the goal of outputting a raster of snowfall across the entirety of Western Montana, utilizing spatial interpolation techniques suits this task. We began by trying Inverse Distance Weighting (IDW), and ordinary Kriging. These methods proved strong in their ease of use, and simplicity of interpretation. However, there is one critical assumption that each of these methods makes that does not suit our analysis. Each method assumes stationarity, which means that a constant mean and variance is assumed across the study area. To combat this assumption, we used an Empirical Bayesian Kriging (EBK) Regression Prediction. EBK can handle non-stationarity by incorporating random processes into the kriging process. We ran EBK on data from 2010 and 2020. The majority of Western Montana, particularly along the border of Montana and Idaho, sees an increase in snowfall between the two years. There are several small pockets of land where a decrease is shown, most prominently near the Kalispell area. Across our study area, snowfall values range from less than 10 inches to over 100. However, due to the limited scope of the analysis, some predicted snowfall amounts may be too heavily weighted by elevation.

1. Problem Statement

Understanding and quantifying—let alone halting—the effects of climate change represents one of the most important issues of our time. Human-caused climate change has been linked to increases in extreme weather, changes in yearly temperature patterns, and even more specific issues such as quickened infrastructure collapse (Hitz et al., 2004; Palou and Mahmoud, 2019). It is safe to say that, whether or not public opinion places climate change at the forefront of issues, it needs to be studied and addressed with the full backing of academia. The problem gets deeper when we consider changes in Geography. Differences in latitude, elevation, physical features, and human-built infrastructure all contribute to specific differences in climate (UCAR, 2024). It is necessary to geographically focus our research to mitigate the specific regional effects of climate change and build better locational resilience.

One such area is the Rocky Mountain West, which geographical region stretches from northern British Columbia, Canada down to the U.S. Southwest. It is primarily characterized by its namesake, the Rocky Mountain Range. Its climate is mostly highland and semiarid, making water the most important resource throughout the year (Running, 2009). The winter season can be especially harsh with historically high volumes of snow and cold temperatures. This physical geography has created a very careful and interconnected ecosystem. Yearly snowfall is critical for the region as the snowpacks that remain in the peaks throughout the summer months represent the region's yearly water supply through meltwater to lakes, tributaries, and rivers—all of which stand to be disrupted by Climate Change (Gergel et al., 2017).

The specific forms of change predicted for the climate of the Rocky Mountain West are "rising temperatures, less snow and more rain, less water stored in snowpack, earlier spring snowmelt and peak runoff, and lower stream flows in summer" (Running, 2009). These changes will have numerous deleterious effects including "increased drought, wildfires, water stress, insect infestations, and economic fall out" (Running, 2009). Changes in temperature and snowfall are at the center of these issues. Rising temperatures will cause less snow in the winter, reduced water reserves for the rest of the year, and faster runoff in the spring, leading to spring floods that can reshape or destabilize river beds. This has already begun to happen; historic flooding in Western Montana in 2022 led to destruction across the watershed and a declaration of Major Disaster from the President (FEMA, 2022).

As these issues continue to proliferate, it is necessary to attempt to understand and project these changes to improve our reactions and resilience to climate change. One such way to do this is to predict the most important factor within this complex system: snowfall. By predicting the snowfall depth across the Rocky Mountain Region based on available data, we can prepare for changes throughout the warm months such as drought or flooding. This project seeks to aid in this process by predicting snowfall in Western Montana (**Figure 1**), a representative area of the Rocky Mountain West with large variations in elevation (**Figure 2**) that also sits right along the continental divide, an important area for understanding water across the entire continent. As it stands right now, most snowfall data is provided from individual weather stations across the state. In order to predict snowfall across the state, we must extrapolate from these points while considering changes in physical geography. Our problem then becomes: How can we accurately predict snowfall across western Montana from only weather station point data?

#	Requirement	Defined As	Data (Spatial)	Datasets Used
1	Data Acquisition	Locate the necessary data for the analysis: NWCC, Montana.gov, and Montana DNRC.	Vector & Raster	Montana Counties, Weather Stations, Montana DEM
2	Define Project Extent	A Polygon layer of counties on or west of the Rocky Mountains in Montana was created to define the extent of the analysis.	Vector	Montana Counties
3	Prepare Vector Data	Specify the parameters for weather station data: stations in Montana, total snowfall, month of February.	Vector	Weather Stations: Total snowfall
4	Prepare DEM	Add Montana DEM to the map, then resample the output cell size so the data is easier to work with.	Raster	Montana DEM
5	IDW	Interpolate raster surface points for total snowfall using Inverse Distance Weights. Compare results with steps 6 and 7.	Raster	Weather Stations -> Predicted Snowfall
6	Kriging	Interpolate raster surface points for total snowfall using Kriging. Compare results with steps 5 and 7.	Raster	Weather Stations -> Predicted Snowfall
7	EBK Regression Prediction	Interpolate raster surface points for total snowfall using EBK Regression Prediction with the Montana DEM as a prediction raster. Compare results with steps 5 and 6.	Geospatial Analysis Layer	Weather Stations + Montana DEM -> Predicted Snowfall
8	Replicate Processing Methods	Select the best procedure out of steps 5-7 and repeat that process for all of the years being analyzed.	Raster	Predicted Snowfall for ~~1990~~, ~~2000~~, 2010, and 2020
9	Compare Results Across Years	Use the Compute Change Raster tool to compare between prediction years by the difference in pixel values.	Raster	Difference of Predicted Snowfall

Table 1: Project requirements

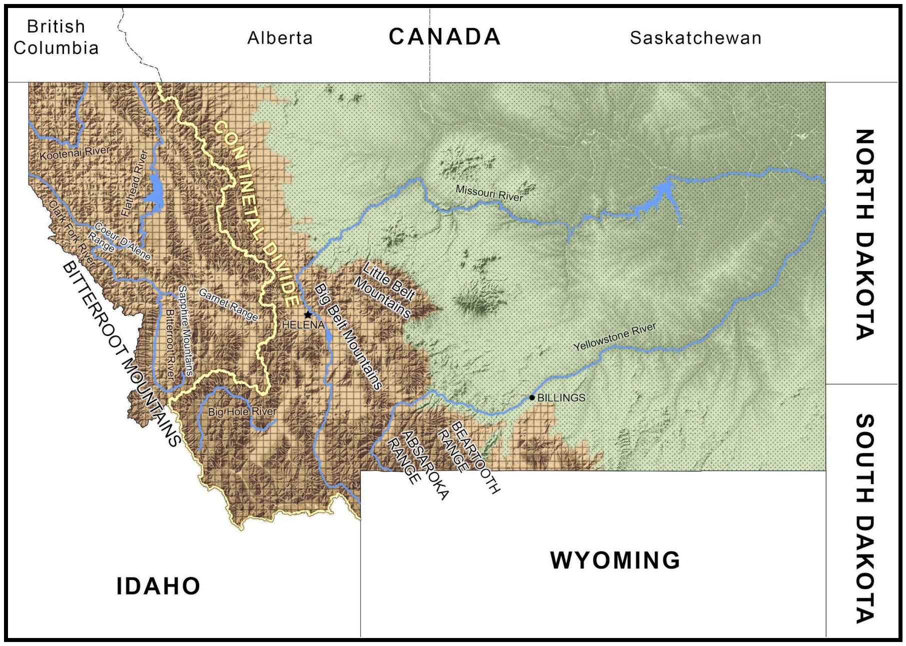

Figure 1: The State of Montana featuring the Rocky Mountain Region and the Continental Divide in Brown.

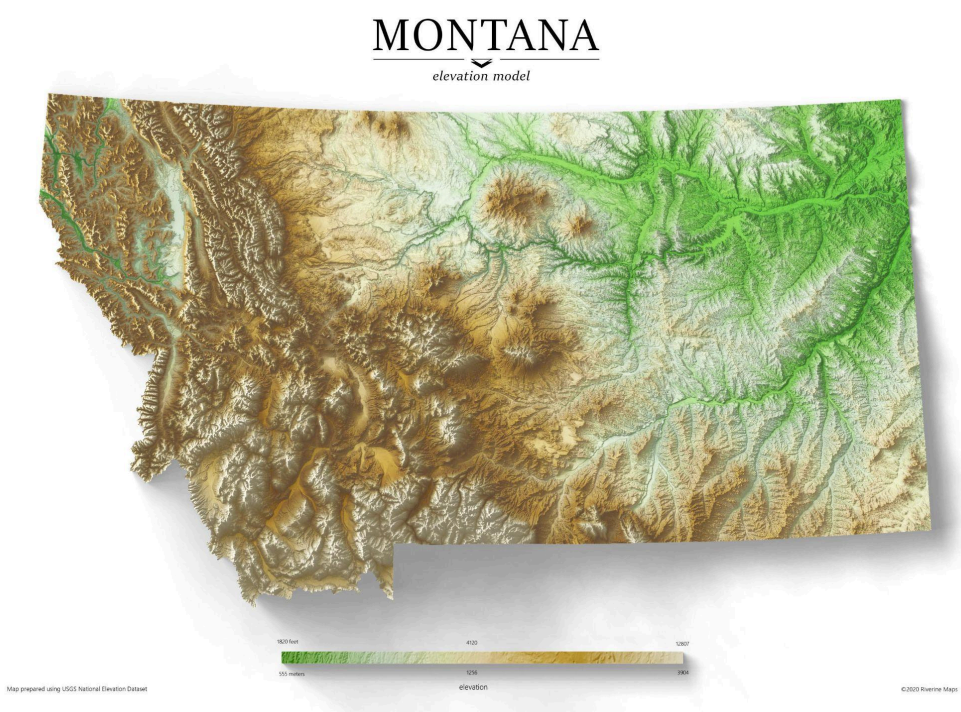

Figure 2: Elevation Map of Montana made by RiverineMaps

2. Input Data

#	Title	Purpose	Spatial Data Type	CRS	Source
1	Montana Weather Stations	Provides historical snowfall data as well as central points for interpolation/regression	Vector: .xml to Points	Projected to NAD 83 State Plane Montana	National Water and Climate Center, 2024
2	Montana Counties Shapefile	Provides spatial extent: Montana counties on or west of the Rocky Mountains	Vector: Polygons	NAD 83 State Plane Montana	Montana.gov, 2016
3	Montana DEM	Provides elevation data that will allow for a more accurate snowfall prediction	Raster Dataset	NAD 83 State Plane Montana	Montana DNRC, 2021

Table 2: Data sources

3. Methods

3.1 Exploratory Analysis

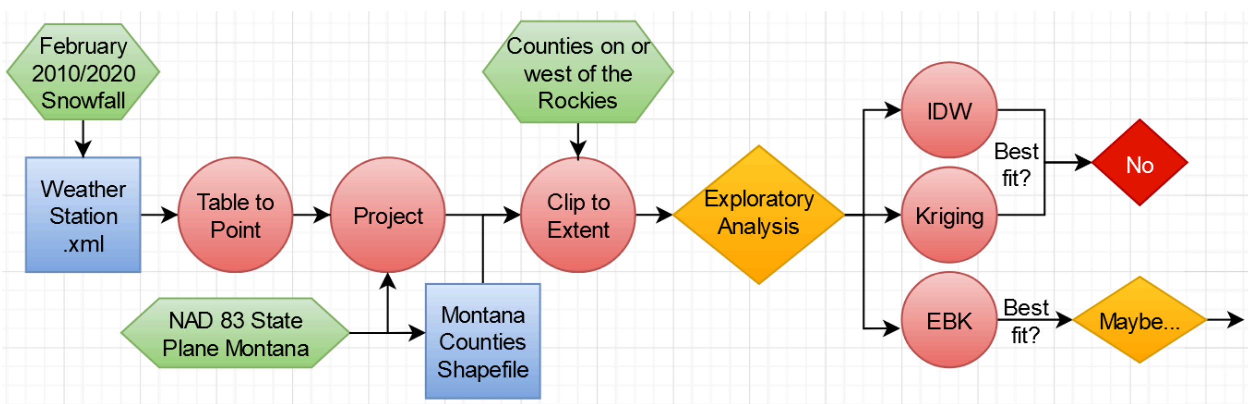

Figure 3: Exploratory analysis workflow

Before interpolation can occur, the input data and temporal extent of the analysis must be determined. The National Water and Climate Center (NWCC) has an interactive map that contains precipitation data for weather stations in western North America—including Montana. Total snowfall can be evaluated on a daily or monthly basis, and data can be retrieved from as far back as 50 years ago. We chose to analyze February snowfall data from 1990, 2000, 2010, and 2020. This would allow us to examine the height of winter snowfall at four recent decade intervals. However, due to later data errors with the 1990 and 2000 data, we ultimately analyzed just February 2010 and 2020 snowfall. After clipping the Montana weather stations to counties on or west of the Rocky Mountains, we were left with more than 150 points of data for our interpolation. These points and county outlines were then projected to NAD 83 State Plane Montana—Montana's singular coordinate reference system.

For exploratory analysis, we explored three interpolation methods: Inverse Distance Weighted (IDW), Kriging, and Empirical Bayesian Kriging (EBK). IDW and Kriging are the two most common interpolation methods, and thus a suitable place to begin our analysis. The simple premise behind IDW is that if interpolating point *i,* data points closer to point *i* will be given greater weight than those farther away. This all depends on the distance power function. The distance away from point *i* to any given point *j* is raised to the power ϱ. If ϱ is 0, this would indicate that with an increase in distance, the weight the snowfall value will get will be the same across all points. ArcGIS Pro uses a default power value ϱ of 2, which is what we used for this analysis.

Kriging is the next step up in terms of interpolation techniques. Unlike IDW, which directly relies on the surrounding points and their values, Kriging utilizes statistical models to help describe the relationship between points. First, we need to find the spatial autocorrelation. The autocorrelation is used in conjunction with the distance that point *i* is away from point *j* to determine the weight that each snowfall value is given. To find spatial autocorrelation, the semivariance between pairs of points is found. The semivariance refers to calculating the difference squared between snowfall values for pairs of points. At each distance *h*, the semivariance of all point pairs with distance *h* is calculated. The result is a plot called a semivariogram, which has all the semi-variances plotted on the y axis and the distances on the x axis. We then fit a statistical model to the semivariogram to help explain the spatial distribution. We chose an exponential model to do this. Then, with the weights having been calculated using the modeled semivariogram and distances, we can interpolate at unknown locations. Kriging provides a more sophisticated approach to interpolation because spatial autocorrelation is incorporated into the process. Examples of IDW and Kriging interpolations are shown below in **Figure 4**.

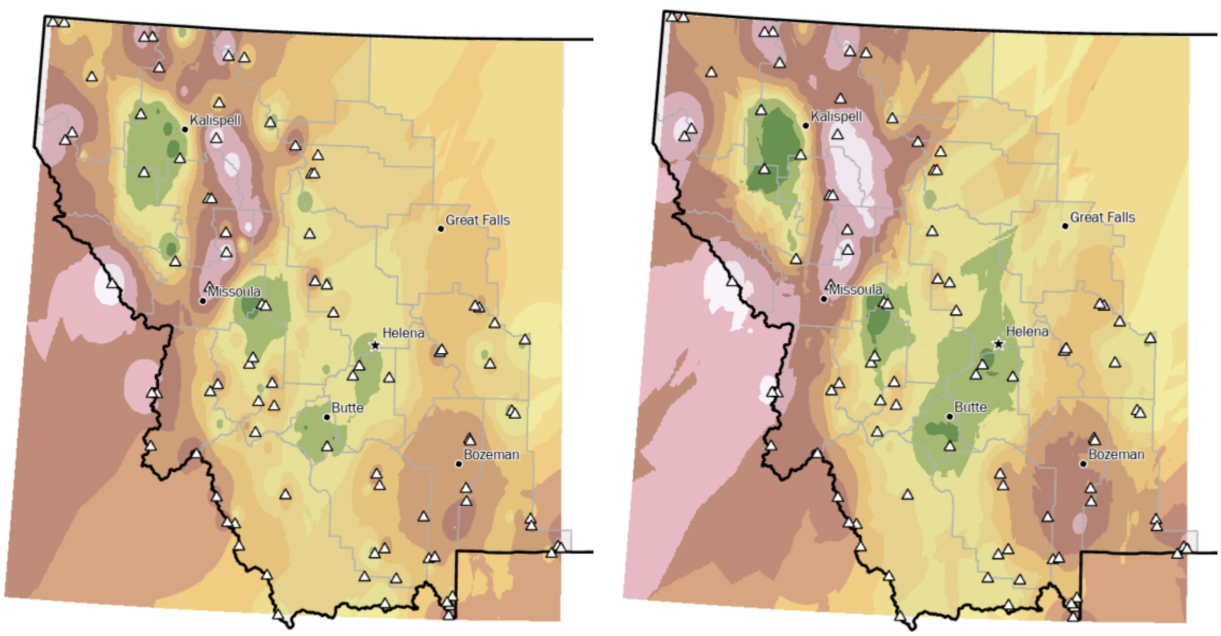

Figure 4: IDW (left) and exponential-model Kriging (right) interpolations of February 2020 snowfall in Western Montana

The final method we tried was Empirical Bayesian Kriging (EBK). EBK differs from ordinary Kriging by utilizing random processes to more accurately model non-stationary data.

Subsets of the data are used, and semivariograms are calculated at each subset. Then, new data is simulated at each of the input locations in the subset from the semivariogram. With this new data, a new semivariogram is calculated. This process repeats a number of times (the default setting is 100 times, which we used). The resulting semivariograms are then merged together in a way that allows us to utilize the new information gleaned from this process. The big benefit of using these random processes is that EBK can handle moderately non-stationary data. Because of the varied topography of Western Montana, assuming stationarity—which is an assumption that both IDW and ordinary Kriging make—is an assumption that we are not willing to make. Using EBK allows us to still interpolate with our non-stationary data.

However, none of these interpolations were the best fit for our analysis; they could not factor in elevation using the Montana DEM. While EBK can handle moderately non-stationary data, we still aren't able to explicitly include elevation as a predictor or part of our analysis. After all, it is elevation that is the main cause of non-stationarity in our study area. Kriging and IDW assume a constant mean throughout the study area, and certainly don't include any covariates such as elevation. As explained above, this makes these methods invalid for our data. There exists some regional variation in mean snowfall across Western Montana that wouldn't be accounted for had we used IDW, Kriging, or EBK without regression.

3.2 EBK Regression Prediction and Comparison

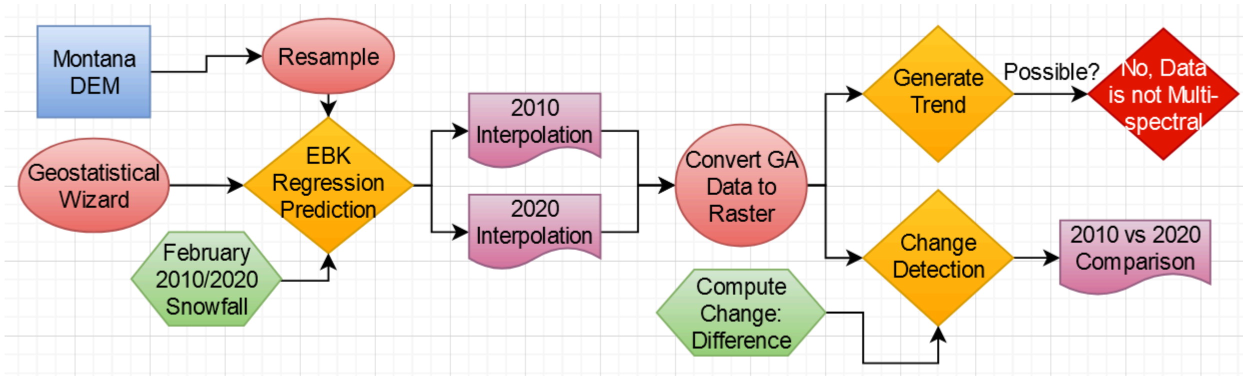

Figure 5: EBK Regression Prediction and Comparison workflows

Although initial tests with EBK were insufficient, ArcGIS Pro's Geostatistical Wizard allowed us to run an EBK Regression Prediction. This interpolation allowed us to use the Montana DEM as an input explanatory variable—far more accurate than just an array of points with snowfall values. By combining regression and EBK, we'll get a more accurate prediction than we otherwise would have. After resampling the Montana DEM to 1000 by 1000 cells, an EBK Regression Prediction was performed for both February 2010 and 2020 snowfall data.

From here, we wanted to generate a trend to possibly predict snowfall in 2030. However, even after converting the EBK Regression Prediction's geospatial analysis data (GA data) to a raster, a trend could not be generated. This is because the data contains only one band of raster data. Trends can only be generated with multispectral data, which would be possible if, for example, multiple months in the same year were analyzed. As a result, we chose to perform a pixel-differential change detection: an analysis that can be performed with two single-band rasters. Both of these analyses—EBK Regression Prediction and Change Detection—are shown in **Figure 6** and **Figure 7**.

4. Results

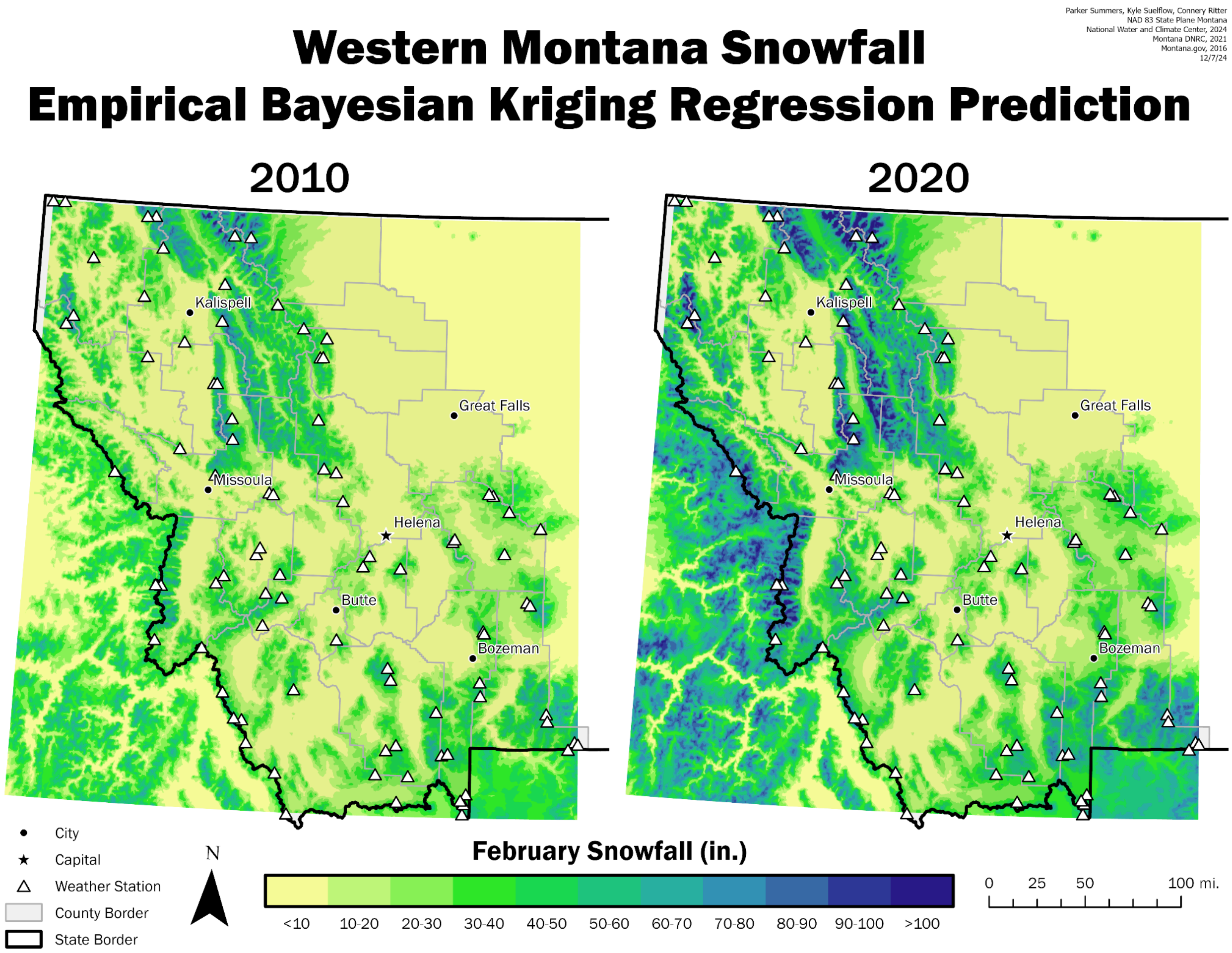

Figure 6: February snowfall predictions for Western Montana in 2010 and 2020 using Empirical Bayesian Kriging Regression Prediction.

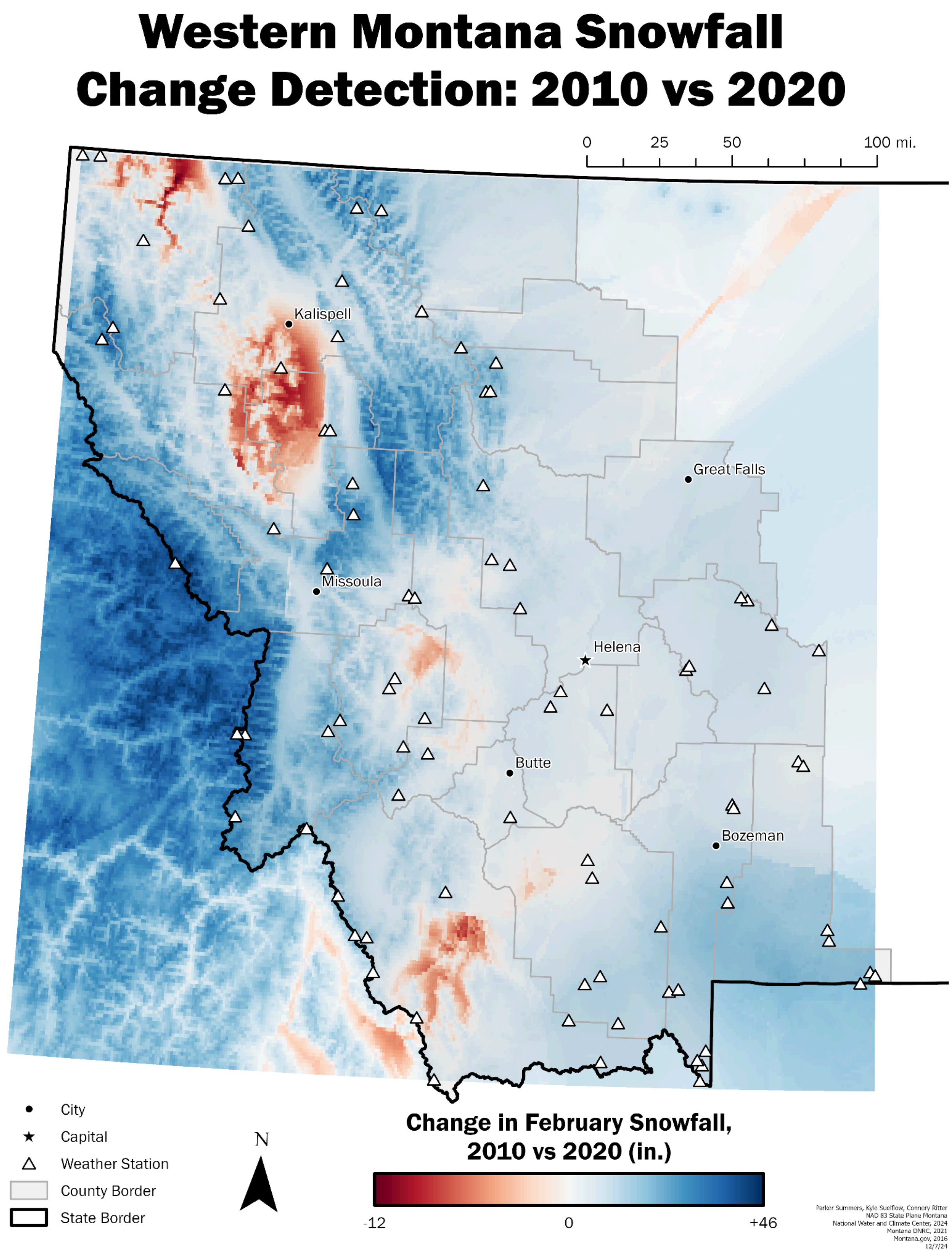

Figure 7: Change in predicted February snowfall for Western Montana from 2010 to 2020.

5. Discussion

We faced a fair amount of limiting questions when addressing our question. The first came with the NWCC website itself. We were not able to get a data range for larger than a month at a time, which limited our prediction to the month of February, as it has higher snowfall. Along with that, we originally wanted to study just one county in Montana, but there was not a sufficient number of weather stations to perform such localized analysis. We decided to pick Western Montana as it provided enough sample points, but remained a relatively small and sufficiently mountainous area. We also needed to account for the large changes in elevation across our chosen spatial range. This meant that we had to carefully select our exploratory analysis methods, as anything that didn't account for such an important variable would not be accurate. We eventually settled on our three—IDW, Kriging, and EBK (later, EBK Regression Prediction)—as they stood to be more accurate. EBK Regression Prediction ended up being the best of the three since it could utilize the Montana DEM as a prediction raster while the others could not. Though we again ran into data trouble when we learned that we could not use much of the raster comparison tools without multivariate raster data.

A broad insight gained from our data analysis is that across Western Montana, changes in snowfall have not been uniform. The southern portion of the state, which tends to be higher elevation (especially closer to the border with Wyoming), largely did not see much change. This stands in contrast to the northern half of our research area which has our highest and lowest snowfall changes between 2010 and 2020. The distribution of weather stations is uniform enough that we feel that there is a real difference in these areas. We hypothesize that the larger variances in elevation in the north might increase the variance in snowfall. This could especially explain the large red and blue sections next to each other in **Figure 7**. If we compare these to the elevation map in **Figure 2** there is strong overlap. This does make it necessary to question if our model is accounting too much for elevation. It would be good to test this phenomenon with more data points from different years.

A somewhat more speculative insight from our data has to do with forest fire occurrences in Montana in the years around 2020. The area of Flathead County has had multiple record breaking fire years since 2017 when a massive fire almost destroyed the town of Seeley. Interestingly, this is the same area covered by the starkest decreases in snowfall on our 2010 and 2020 comparison map. More specifically, there was a large fire in 2020 on the Flathead Reservation, which overlaps significantly with the specific area of snow loss on our map (Dresser, 2020). Along with this correlation, there was also a large wildfire in the Beaverhead mountains along the Montana-Idaho border, again overlapping the large area of decrease on our comparison map (Murray, 2020). Although it is impossible to prove this linkage with our data statistically, it does provide an interesting insight. There were plenty of other wildfires in the year 2020 that have no indication of a snowfall decrease, such as around Helena and Bozeman which both saw fires that year (Murray, 2020). If we had more data over more time, it would be very interesting to investigate these overlaps further and see if more wildfires have these overlaps or if this is just a coincidence.

Given another three months to work on this analysis, we would run EBK Regression Predictions on other months or years in Western Montana to create a prediction for the future. For example, every month with snowfall in 2020 (November to April), or a five-year span ranging from 2015 to 2020 in February could be predicted and compared. This data would likely consist of multispectral data, so more advanced forms of change detection can be carried out, such as generating a trend for future snowfall. Further analysis would help alleviate the limited scope of our current analysis; comparing two years alone cannot generate a trend. This would also allow us to assess whether elevation plays too much of a role in snowfall interpolation.

In this analysis, we sought to predict snowfall using only weather station point data. We were able to successfully predict February 2010 and February 2020 snowfall across Western Montana using EBK Regression Prediction and compare the changes between the two. This analysis gave us valuable insight into interpolation methods like IDW and Kriging and allowed us to utilize them practically while understanding their inherent limitations. Although our original goals were constricted by both data and analytical obstacles, we were able to create a workable and adaptable model that can be reproduced across the region. By confronting such a complex issue through straightforward, step-by-step analysis, we can begin to take on larger pieces and problems using our previous expertise—all while being ready to confront new issues through measured experimentation and critical thinking.

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